Atlantic sea scallop (*Placopecten magellanicus*)

Summary Scores

- **Sensitivity:** High
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Range			Georges Bank					
	Score	LH DQ	Tally	Score	LH DQ	Tally	Score	LH DQ	Tally
Adult Mobility	3.84	2.73		3.84	2.73		3.84	2.73	
Complexity in Reproductive Strategy	2.14	2.95		2.12	2.95		2.16	2.95	
Dispersal of Early Life Stages	1.8	3		1.8	3		1.8	3	
Egg and Larval Survival and Settlement Requirements	2.28	2.95		2.28	2.95		2.28	2.95	
Historical Variability in Temperature	1.64	2.9		1.64	2.9		1.64	2.9	
Juvenile Habitat Requirements	1.12	3		1.12	3		1.12	3	
Other Stressors	2.54	2.98		2.48	2.98		2.6	2.98	
Population Growth Rate	2.34	2.78		2.32	2.78		2.36	2.78	
Prey Specificity	1.16	3		1.16	3		1.16	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	3.96	2.2		3.96	2.2		3.96	2.2	
Spawning Cycle	2.28	3		2.28	3		2.28	3	
Stock Size/Status	1.8	3		1.88	3		1.72	3	
Total Sensitivity | Certainty	**High (3) | 1**			**High (3) | 1**			**High (3) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.6	1.10	
Bottom Oxygen	2.88	0.99	
Bottom Salinity	2.83	0.91	
Bottom Aragonite Solubility	1.22	0.67	
Net primary production	1.06	0.17	
Downward Flux of Particulate Organic Carbon	1.27	0.33	
Total Exposure	**Moderate (2.21)**	**0.92**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (6.62)**	**2.77**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Atlantic sea scallop. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High	High	None
Exposure	High	Moderate	Decrease
Vulnerability	High	Moderate	Decrease
Directional Effect	Negative	Negative	None

Stocks

Atlantic sea scallop is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

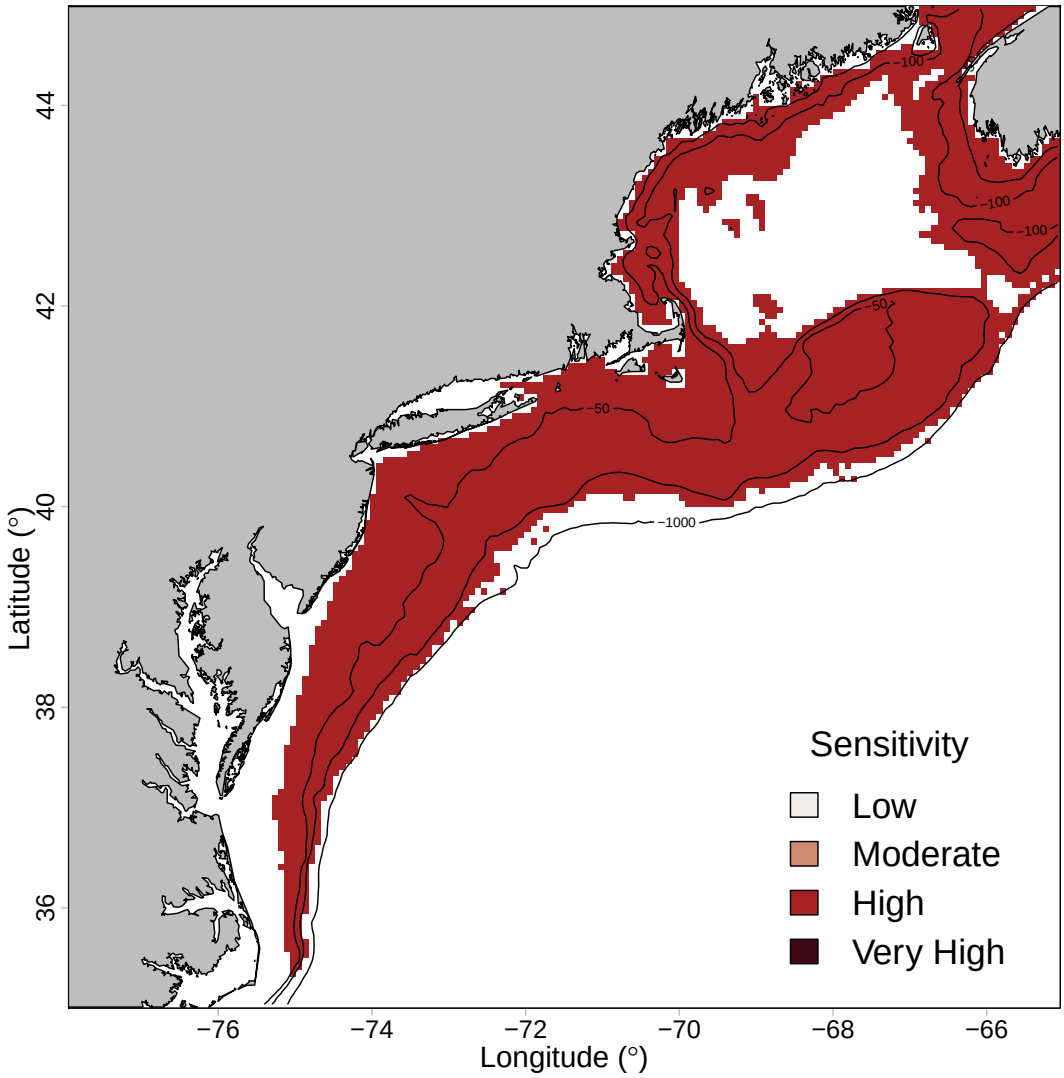


Figure 2: Atlantic sea scallop Sensitivity on the Northeast Shelf

Both the Georges Bank and Mid Atlantic Bight stock units of Atlantic Sea Scallop received a **High** sensitivity score (Figure 1). Certainties for both stocks were 100%. Relationship to Calcified Organisms (4) and Adult Mobility (3.8) drove the scores in both stocks. Scores for each attribute were the same for both stocks (Figure 1).

Scallops are subject to stress due to increasing ocean acidification due to their calcium carbonate shells (Zang et al. 2022). Adults are primarily sessile, but can make small-scale movements to avoid predators (Hart and Chute 2004).

Atlantic sea scallops also received a High sensitivity score in the previous assessment (Hare et al. 2016). This score was also driven by Sensitivity to Ocean Acidification and Adult Mobility (Hare et al. 2016).

Exposure:

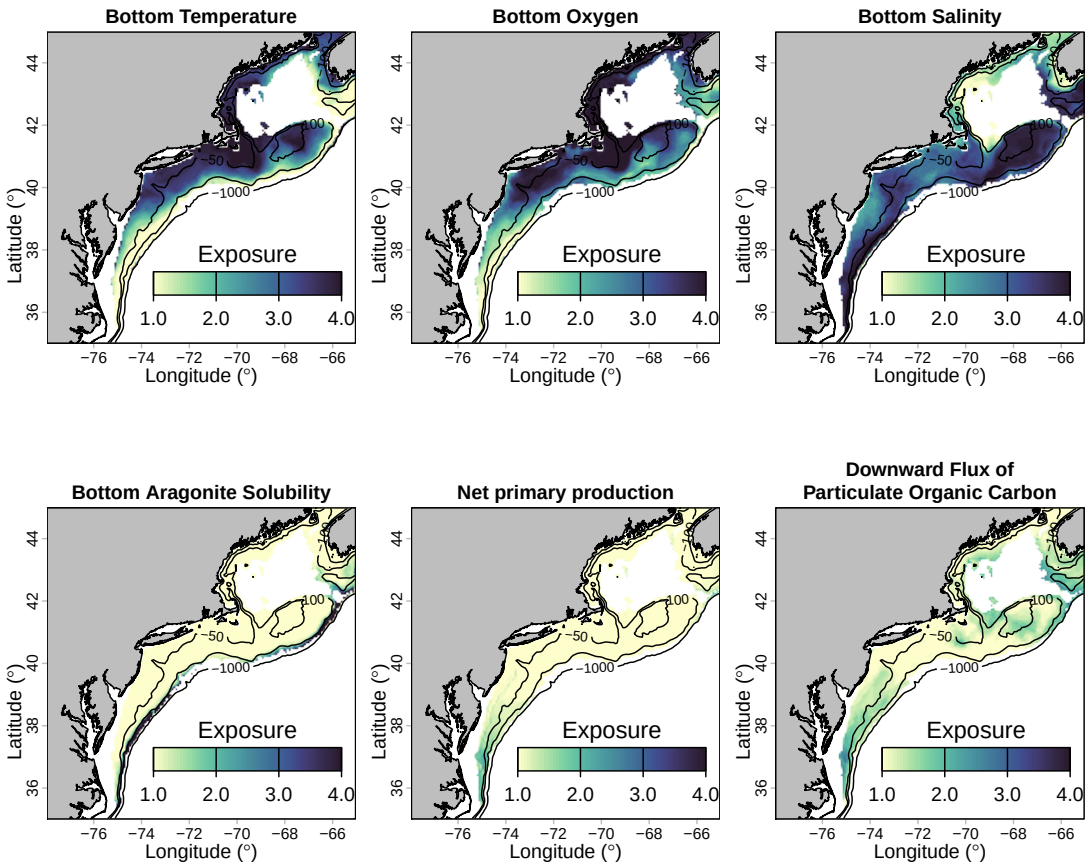


Figure 3: Maps of each Exposure Factor for Atlantic sea scallop. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for Atlantic Sea Scallop across its range was **Moderate** (2.21 +/- 0.92) (Figure 1). Exposure was Very High over Georges Bank and High in southern New England and along the coast of the Gulf of Maine (Figure 4). These patterns were driven by bottom temperature, salinity, and oxygen exposures (Figure 3).

In the previous Assessment, Atlantic sea scallop received a High exposure score, driven by sea surface temperature and ocean acidification exposure (Hare et al. 2016).

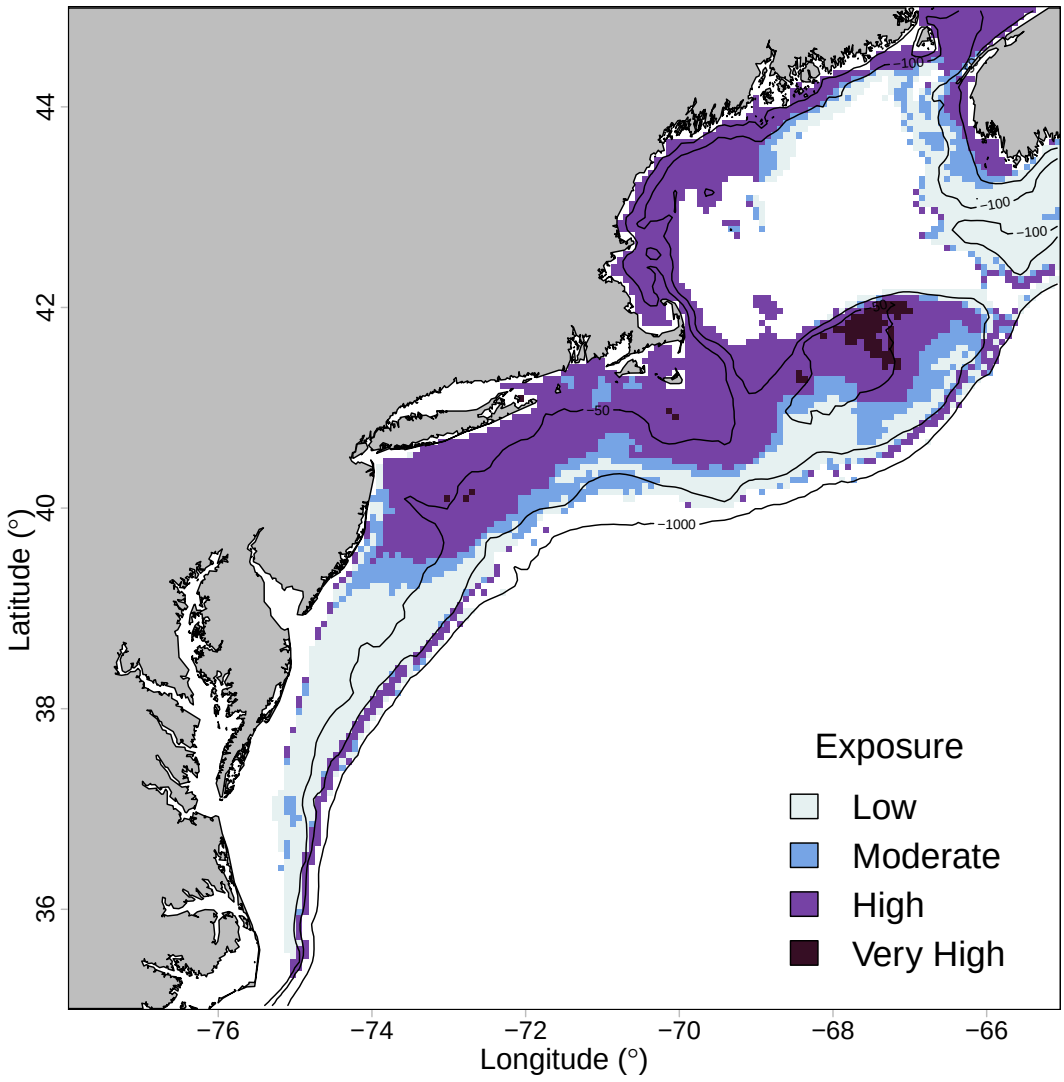


Figure 4: Total Exposure on the Northeast Shelf for Atlantic sea scallop. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

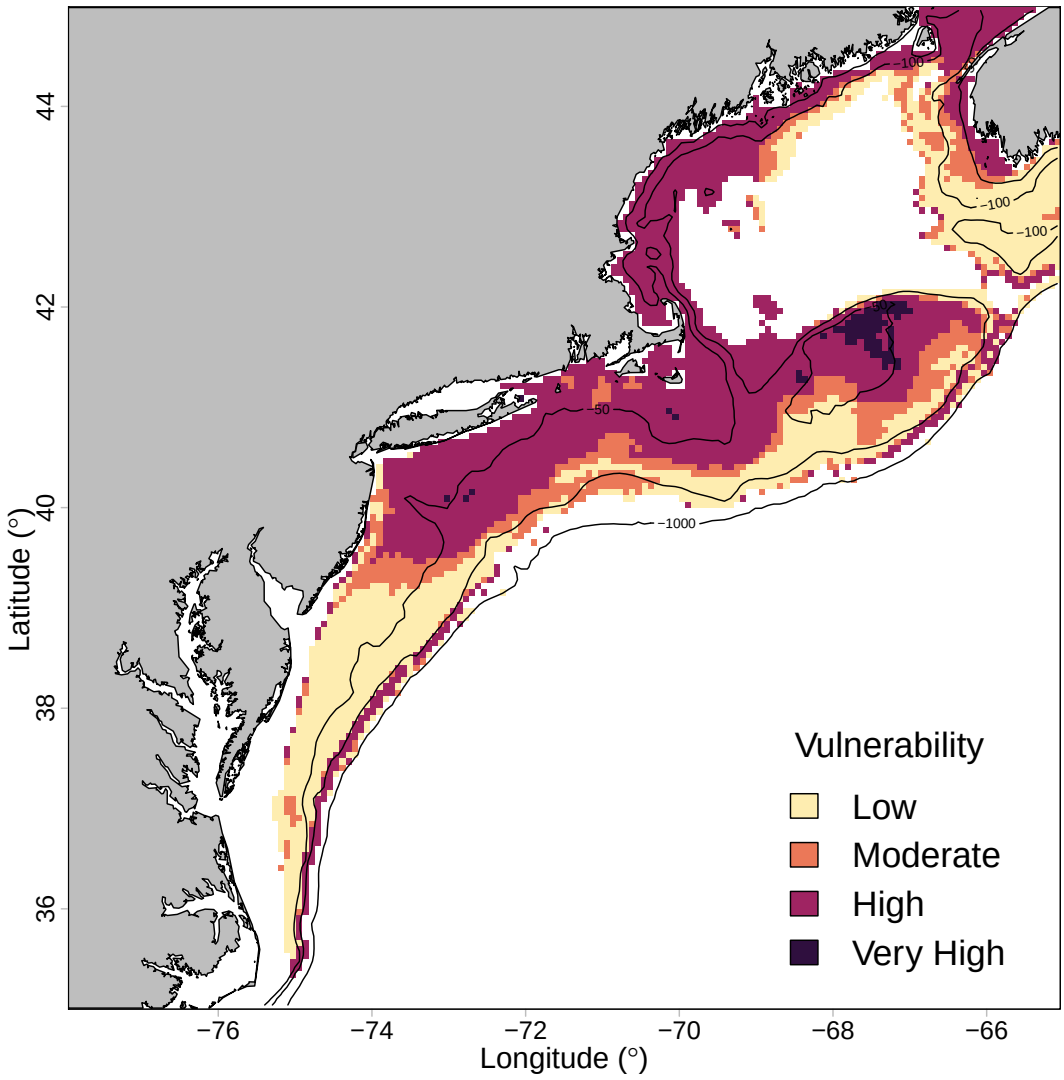


Figure 5: Vulnerability on the Northeast Shelf for Atlantic sea scallop. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Atlantic Sea Scallop across its range was **Moderate** (6.62 +/- 2.77) (Figure 1). Vulnerability was Very High over Georges Bank and High throughout the coast of the Gulf of Maine and Southern New England (Figure 5).

In the previous Assessment, Atlantic Sea Scallop received a High Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

95% of sensitivity attribute data quality scores were greater than or equal to 2 for both stocks of Atlantic Sea Scallop, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were

found in either stock (Figure 1).

Model Confidence Score:

Mean model confidence scores were 2.6 (+/- 0.5) indicating high confidence in the model. Experts stated that important variables matched their expectations. The ensemble model's AUC was 0.85, indicating good performance.

Additional Metrics

Distribution Shifts:

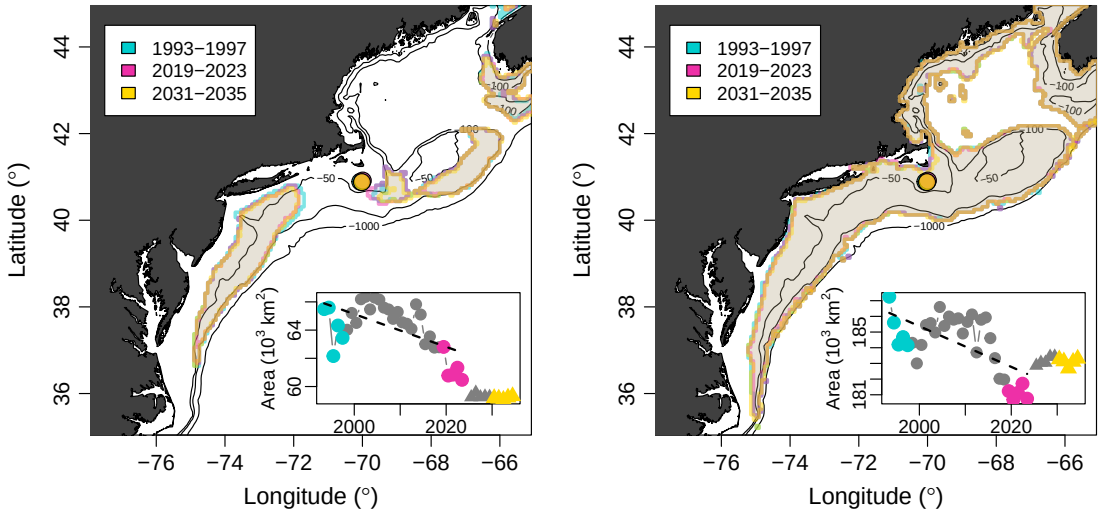


Figure 6: Distribution shifts on the Northeast Shelf for Atlantic sea scallop. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

There were significant declines in sea scallop core habitat area and range between 1993 and 2023. Most core habitat loss occurred in the Mid Atlantic Bight (Figure 6). The weighted centroid shifted less than 10 km east-northeast.

Atlantic sea scallop distributions are expected to remain similar in the next decade. No significant changes in habitat area are expected and the weighted centroid of the distribution is predicted to shift less than five kilometers southwest (Figure 6).

Sea scallops received a Moderate Distributional Vulnerability Rank in the previous Assessment (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	64,498	185,605	6.30	8.90
2019-2023	6.43 km ENE	61,364 (-3,134)	181,195 (-4,410)	6.17 (-0.12)	8.90 (0)
2031-2035	4.43 km SW	59,177 (-2,187)	183,116 (+1,921)	6.30 (+0.12)	9.0 (+0.08)

Directional Effect:

Both stocks of Atlantic sea scallop are estimated to be **negatively** impacted by future climate change. Certainty ranged from 88 – 99%, with higher certainty for the Mid Atlantic Bight stock. Rising temperatures and ocean acidification induce stress in sea scallops. In addition, anthropogenic activities such as dredging and offshore wind construction degrade habitats and impact behaviors. These results are similar to those from the 2016 Climate Vulnerability Assessment (Hare et al. 2016).

Climate Effects on Abundance and Distribution

The Atlantic sea scallop population in the Northwest Atlantic is subject to climate-related environmental stress from rising temperatures and ocean acidification (Zang et al. 2022). Scallop habitat can be degraded by trawling/dredging activity (Maguire et al. 2002). Scallop feeding behaviors are temporarily impacted by noise pollution associated with offshore wind construction (J'ez'equel et al. 2022). Changes in temperature and food availability are likely the predominant drivers for reduced growth in sea scallops (Cranford et al. 1998). Rheuban et al. (2018) modeled the Atlantic sea scallop fishery under a range of climate and management scenarios and found the possibility of reduced biomass up to 50% by the end of this century. Results of a scope for growth model with increased warming food deficiency indicated a range contraction of suitable sea scallop habitat in the Mid-Atlantic Bight (Zang et al. 2022). A second scope for growth model by Zang et al. (2023) investigated the relationship between sea scallop size and temperature. Under future climate warming, sea scallops exhibited a range contraction and decline in maximum shell height. An integrated assessment model simulating sea scallop population dynamics with oceanographic and socioeconomic influences indicates that sea scallop harvests could experience a substantial decline by 2050 under increased CO^2 emissions (with the assumption that ocean acidification decreases recruitment and slows growth and warming increases growth) (Cooley et al. 2015).

Life History Synopsis

Atlantic sea scallops are one of the most economically important fisheries in the Northwest Atlantic, with average annual ex-vessel revenues of over $500 million (Zang et al. 2022). Their habitat ranges from the Gulf of St. Lawrence to Cape Hatteras, North Carolina (Stewart and Arnold 1994) with primary fishing areas in Georges Bank, the Great South Channel, and the Mid-Atlantic Bight (Hart and Rago 2006). After a period of severe overfishing in the 1970s to 1990s, regulations including effort reductions, gear restrictions, and closed areas were introduced to rebuild the stock (Hart and Rago 2006). Spawning occurs in late summer or early fall, although smaller spawning events at other times of the year are possible

(Culliney 1974). Gametes from other scallops, agitation, or tides may trigger scallop spawning (Parsons et al. 1992). Larvae reside in the water column for 5-7 weeks (Thouzeau, Robert, and Smith 1991) and circulate with residual currents ranging from 6-25 cm/s (Culliney 1974). Larval exchange is influenced by duration and depth of planktonic drift, gyre strength, and weak cross-isobath flow (Thouzeau, Robert, and Smith 1991). The transition to the benthos of the freeswimming larvae occurs at a size of about 0.25 mm shell height and is accompanied by drastic changes in diet, morphology, and locomotory ability (Bourne 1965). Mean fall temperatures impact the time larvae take to settle (Tremblay et al. 1994). Sea scallop larvae are potentially preyed on by filter feeders and larval and adult planktivores (Tremblay and Sinclair 1986). Juvenile Atlantic sea scallops are mainly found on gravel, small rocks, shells, and among branching animals and plants that permit attachment of juveniles (Thouzeau, Robert, and Smith 1991) and feed on suspended particles, primarily phytoplankton and microzooplankton (Shumway, Selvin, and Schick 1987). Adult sea scallops are filter feeders on phytoplankton and other suspended organic particulates from the water (Shumway, Selvin, and Schick 1987). Adult sea scallops prefer temperatures of 5-18°C (Hart and Chute 2004). Predation on juvenile and adult sea scallops decreases with shell height; common predators include Atlantic cod, wolffish, sculpins, winter and yellowtail flounder, as well as boring polychaetes, sea stars, crabs, and lobsters (Hart and Chute 2004). The New England Fishery Management Council manages Atlantic sea scallops in federal waters under the Atlantic Sea Scallop Fishery Management Plan. As of the 2020 Management Track Assessment, the stock is not overfished and overfishing is not occurring (Northeast Fisheries Science Center 2020).

Supplemental Scores

Exposure Timeseries

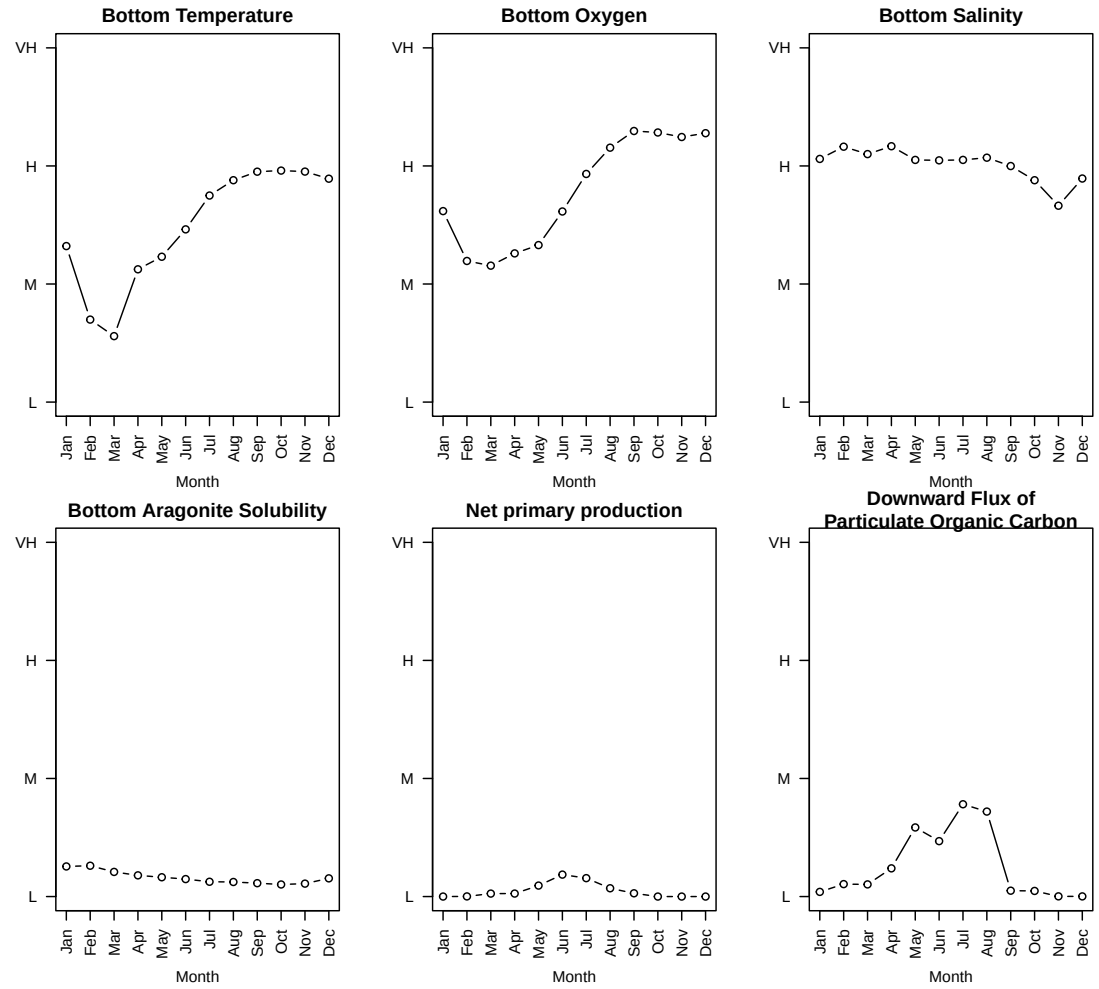


Figure 7: Monthly Timeseries of each Exposure Factor for Atlantic sea scallop. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure ranged from Low to High. Scallops experienced Low exposures in late winter and spring (February through May) (Figure 8). These patterns were driven by changes to bottom temperature and oxygen exposure. Exposure was High in August, driven by increases in downward carbon flux exposure (Figure 7).

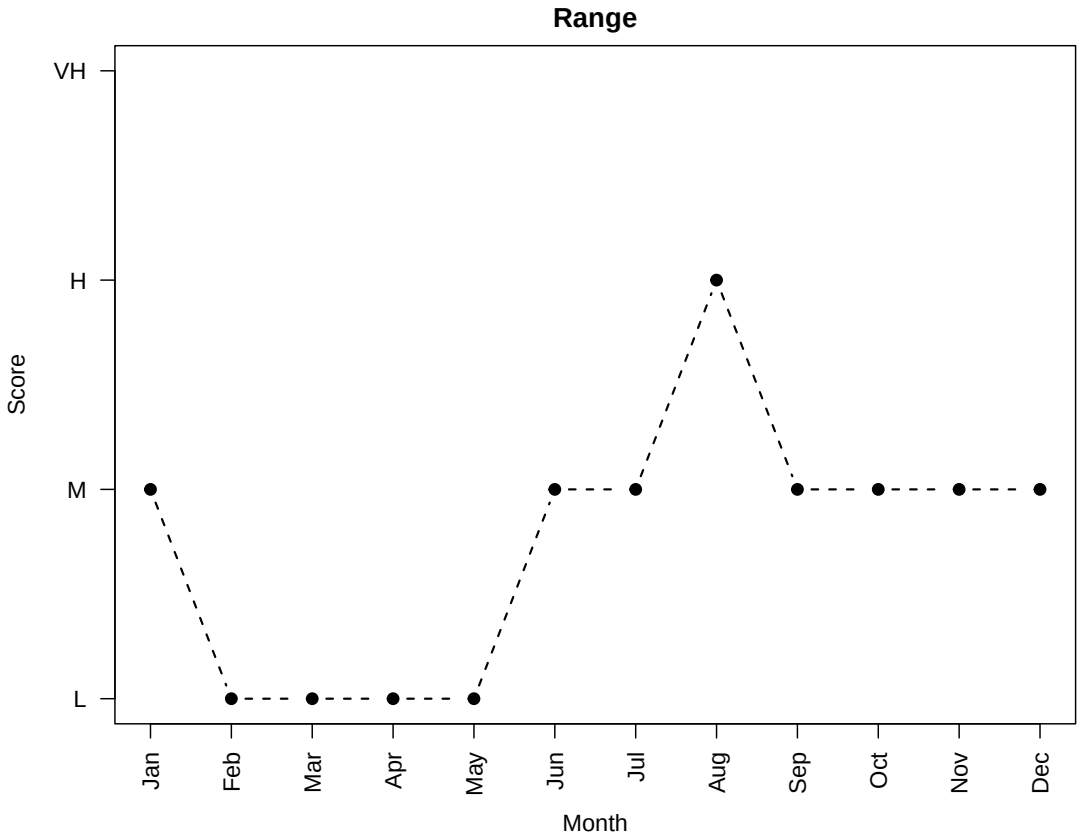


Figure 8: Timeseries of Total Exposure for Atlantic sea scallop

Exposure due to Significant Factors

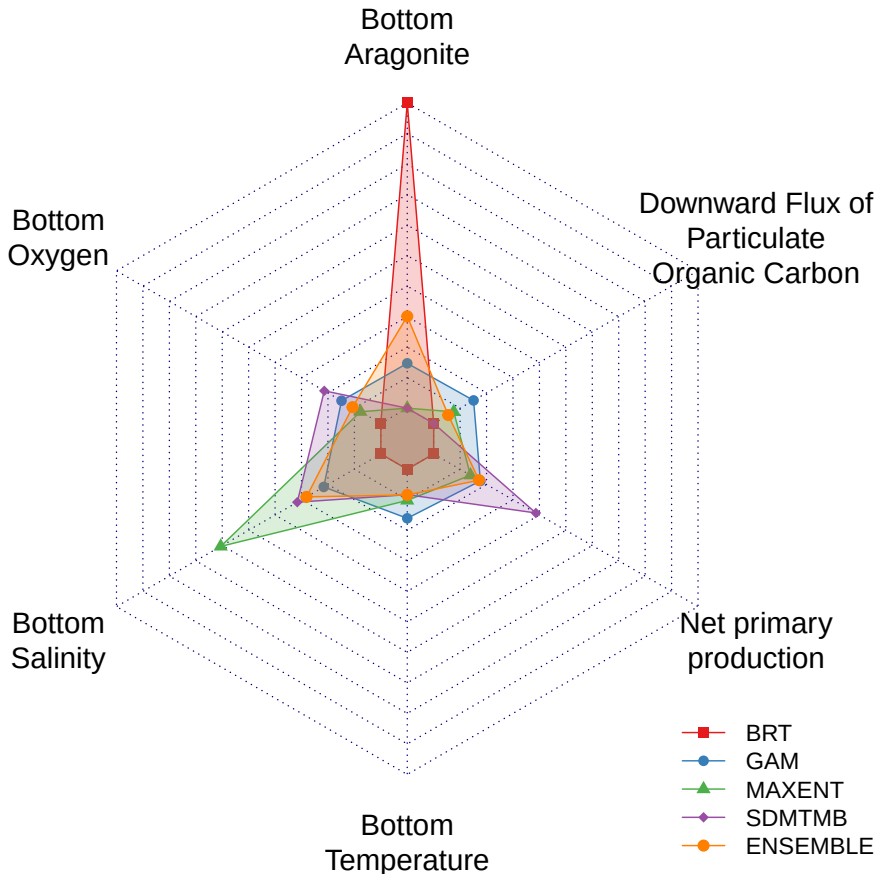


Figure 9: Dynamic Factors driving Atlantic sea scallop distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom oxygen, salinity, and aragonite saturation, in addition to net primary production were important dynamic drivers of sea scallop distribution (Figure 9). Scallops feed on suspended particles (Shumway, Selvin, and Schick 1987) and are often associated with gravel, other shellfish, and rocks (Thouzeau, Robert, and Smith 1991). When only these factors were used to calculate exposure, mean exposure remained Moderate (Figure 10; Figure 11). High exposure was still found on Georges Bank and in Southern New England (Figure 10).

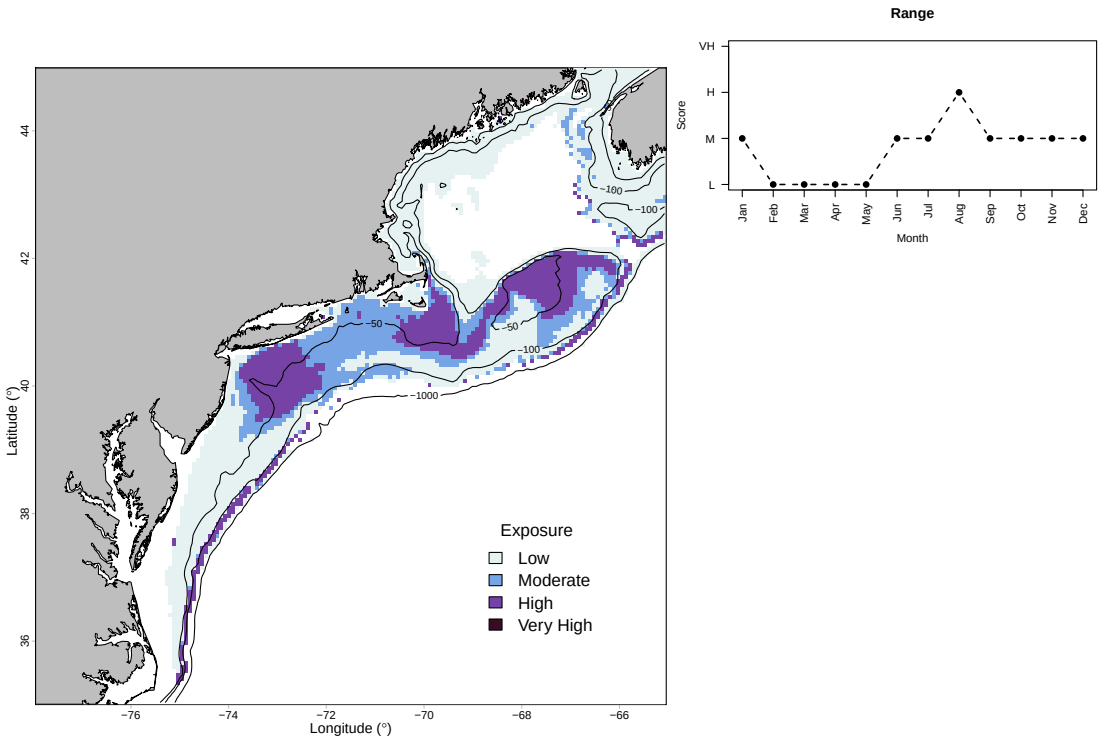


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Atlantic sea scallop distributions

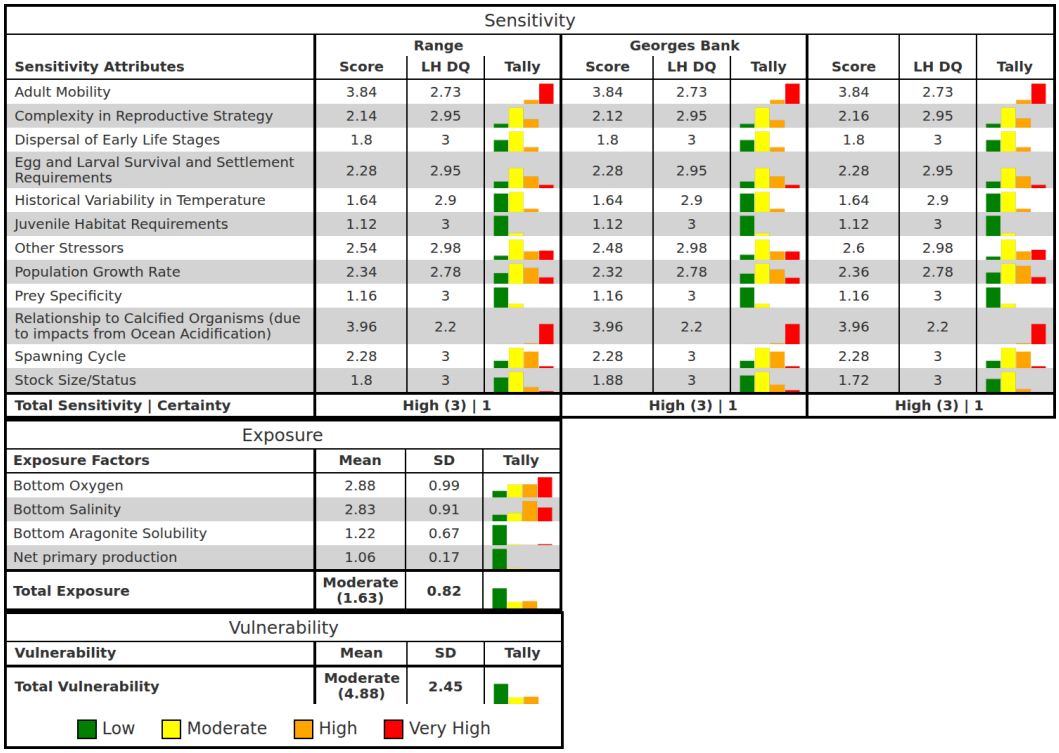


Figure 11: As in Figure 1 for Atlantic sea scallop, but with only significant Exposure factors and the resulting Vulnerability Scores.

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